Universidade Vigo

ANALYSIS OF CONVERSATIONAL DATA FROM HEALTH-FOCUSED CHATBOTS USING NATURAL LANGUAGE PROCESSING TECHNIQUES

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To my parents
To my whole family

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This thesis marks the end of an incredible experience: deciding to come to Vigo to pursue this master's degree. What began as a vague idea of what I wanted to do ended up being the consequence of making a decision without the freedom to fully choose it, in rushed circumstances, yet one I do not regret.

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Glossary

GDPR General Data Protection Regulation; the legal framework for data privacy and protection in the European Union
Synonyms: RGPD . 11, 28

Gold Standard A human-annotated dataset (in this study, 130 sessions) used as a reference to validate the model's performance . 11, 16

Hadamard Product A mathematical operation applied to integrate subtopics into toipcs. It functions as a filtering mechanism that multiplies vectors element-wise to eliminate low-confidence model assignments while preserving the intensity of high-quality self-disclosures. 11, 13, 35, 39

Inference The process of executing a trained model (in this case, Gemma-2-9b-it) to generate predictions or evaluations on new, unseen data based on the provided input or prompt.. 11, 28

Social Penetration Theory The fundamental theoretical framework used to understand how individuals reveal personal information to others. It uses the "onion metaphor" to suggest that personality is structured in various layers (peripheral, intermediate, and core), and that the development of a relationship involves systematically peeling back these layers through communication. 11, 18, 19, 22, 29

Subtopic The fundamental and granular unit of self-disclosure analysis. These are specific themes (e.g., mental health, family dynamics) that provide the raw data for scoring.
Synonyms: Sub-dimensions, Sub-themes, Granular Features.. 11, 13–16

Topic The highest level of semantic abstraction in this study, representing the broad "layers of the onion" in Social Penetration Theory.
Synonyms: Categories, Macro-dimensions, Main Themes, SPT Layers.. 11, 14

Abstract

This Master's Thesis investigates the automated detection of self-disclosure in 16,000 chatbot sessions with elderly users. Grounded in Social Penetration Theory, the study evaluates various feature extraction pipelines using the Gemma-2-9b-it LLM. We compare binary versus rubric-based scoring and direct versus Subtopic-based classification aggregated via Hadamard Product product. An unsupervised K-Means clustering approach is employed to identify thematic patterns. Validation is conducted against a Gold Standard of 130 manually annotated sessions by three experts, utilizing Krippendorff's Alpha. The results provide a robust methodological foundation for monitoring vulnerability and trust-building in conversational AI for senior citizens.

Keywords: Natural Language Processing, Self-Disclosure, LLM, Social Penetration Theory, K-Means Clustering, Elderly Care.

1 Introduction

1.1 Motivation and Context

The progressive aging of the global population has led to a significant increase in social isolation among senior citizens. To address this, AI-driven conversational agents have emerged as a scalable tool for providing companionship and monitoring [1]. However, these interactions pose the risk of reveling personal information, a process known as self-disclosure.

In the field of psichology, understanding how and when a user shares sensitive details (health, memories, or emotions) is critical [2]. This project is motivated by the need to transform unstructured dialogue data from the Serenia tool (taken from a dataset with 16,000 sessions) into useful [3]. By applying Social Penetration Theory (SPT) and Large Language Models (LLMs), we aim to establish the ground for a robust assessment methodology [4]. Consequently, this work addresses the challenge of characterizing LLMs to extract high-level semantic features to detect vulnerabilities and improve the quality of human-machine interaction in sensitive contexts.

1.2 Automated Self-Disclosure Detection

Despite the proliferation of digital tools for elderly care, there is a lack of robust, automated methods to quantify the emotional and personal depth of their interactions. While current Natural Language Processing (NLP) techniques can handle intent recognition or sentiment analysis, their capabilities to capture the nuances of self-disclosure are less understood.

The core challenge lies in the unstructured nature of dialogue data and the subjective complexity of what constitutes a sensitive or deep revelation [5]. Existing approaches have relied for a long time on rigid keyword matching or general-purpose sentiment scores, which are insufficient for identifying the thematic layers (health, relationships, personal history) defined by social psychology [6]. Furthermore, the use of LLMs as evaluators introduces a new problem of methodological reliability: there is no consensus on whether binary classification, multi-level rubrics, or Subtopic aggregation yields the most accurate representation of user trust. Therefore, this research addresses the need for a validated, theory-driven approach to transform raw conversational data into a structured metric of self-disclosure in the context of elderly-AI interaction.

1.3 Social Penetration Theory

SPT provides a foundational framework for understanding how individuals reveal personal information to others [7]. Often illustrated through the onion metaphor, the theory suggests that personality is structured in layers, and relational development involves the systematic peeling of these layers through communication. [8]

According to the theory, the disclosure process is defined by two primary dimensions:

1. **Breadth:** The variety of Topics covered during an interaction.
2. **Depth:** The level of intimacy or sensitivity of the information shared within those topics.

SPT has been widely adapted to study Human-Computer Interaction (HCI). In this thesis, the theory is utilized to categorize conversational data into distinct thematic domains. By mapping user utterances to specific layers (ranging from superficial biographical data to more intimate disclosures regarding *Health*, *Relationships*, and *Personal Development*), we can quantify the level of trust and openness established between the elderly user and the conversational assistant

[9]. This theoretical grounding justifies the selection of our classification categories and provides the necessary psychological context for the subsequent unsupervised learning phase.

1.3.1 Thematic Categorization of Disclosure

In this research, the framework of SPT is adapted to categorize the content of interactions rather than merely measuring the depth of penetration [10]. For the study of elderly-chatbot interactions, SPT domains are divided into three primary thematic categories:

- **Health and Well-being.** This domain encompasses the user's physical, mental, and emotional state. In the context of the elderly, it is a highly sensitive category where disclosure ranges from routine health updates to the expression of chronic concerns or emotional distress.
- **Interpersonal Relationships.** This category focuses on the social and family fabric of the individual. It involves narratives about family dynamics, affective bonds, and social support systems, which are essential for identifying the user's level of social integration or isolation.
- **Personal History and Development.** This thematic area covers the individual's life trajectory, including professional milestones, personal failures, and past traumas. It represents the narrative self, where disclosure often involves significant life events that shape the user's current identity.

The differentiation between these categories is governed by the *norm of reciprocity*, although in Human-Computer Interaction (HCI) this dynamic is altered. The elderly user often perceives the agent as a non-judgmental confidant, which facilitates the transition between neutral topics and highly personal thematic domains [11]. This study focuses on how these specific categories are populated during the interaction, as the presence of disclosure in sensitive areas is a key indicator of the chatbot's ability to create a safe and supportive digital environment.

The situational context and the nature of the communication channel are fundamental to understanding how these categories emerge. As noted in [12], self-disclosure in digital environments is not a linear reflection of relational depth but is driven by functional motivations and technical affordances. The perceived privacy of a one-on-one chatbot interaction reduces the "social cost" of sharing [13]. This allows elderly users to discuss sensitive topics within the specified categories more freely than they would in broader, public digital social settings [14].

1.4 Research Objectives

The primary objective of this thesis is to design and evaluate methods for the automated assessment of self-disclosure in dialogues between elderly users and conversational AI tools, leveraging LLMs and unsupervised learning techniques.

To achieve this overarching goal, the following specific objectives have been established:

O1: Feature Extraction and Scoring. To implement and compare different scoring strategies (specifically binary versus rubric-based (0-5) scales) using a model to extract high-level semantic features from conversational sessions.

O2: Methodological Comparison of Classification. To evaluate the effectiveness of direct category classification versus a bottom-up approach that utilizes Subtopic aggregation through the application of *Hadamard products*.

O3: Unsupervised Pattern Identification. To perform clustering (*K-Means*) on the extracted features to identify thematic domains based on SPT.

O4: Empirical Validation. To validate the model's performance against a human-annotated Gold Standard of 130 sessions, measuring inter-rater reliability via *Krippendorff's Alpha*.

1.5 Work Hypotesis

The present research is built upon the premise that current NLP frameworks for elderly-chatbot interactions often overlook the psychological depth and the temporal evolution of human-machine relationships. By integrating SPT with state-of-the-art GAI, we seek a multidimensional evaluation of emotional intimacy.

The central hypotheses guiding this work are as follows:

- **H1: Semantic Depth through SPT.** The application of SPT to conversational datasets involving elderly users will reveal patterns of self-disclosure that standard NLP metrics fail to capture. This represents a pioneer application of SPT within an NLP framework specifically tailored for the elderly population.
- **H2: Superiority of GAI-ML Hybridization.** A hybrid approach, utilizing LLMs for granular subtopic extraction and traditional Machine Learning for clustering, will provide a more robust and interpretable characterization of user profiles.
- **H3: Resilience of the Gating Mechanism.** The implementation of a Hadamard Product between rubric-based intensity and binary detection will significantly reduce semantic noise, providing a high-fidelity representation of the user's latent disclosure space.
- **H4: Relevance of Non-intrusive Nature.** Unlike traditional systems designed for information extraction, it is desirable to directly analyze the depth of spontaneous self-disclosure in free interactions, prioritizing emotional companionship over explicit health assessment."

1.6 Thesis Structure

This document is structured into seven chapters that guide the reader through the theoretical, technical, and evaluative phases of the research:

- **Chapter 2: State of the Art** explores the psychological foundations of SPT and the technical landscape of LLMs and unsupervised learning.
- **Chapter 3: Methodology** constitutes the technical core of the thesis, describing the experimental study for feature extraction, the application of the *Hadamard product* for Subtopic aggregation, and the *K-Means* clustering configuration.
- **Chapter 4: Results and Validation** assesses LLMs' performance by comparing it against an expert-annotated Gold Standard, using metrics such as *Krippendorff's Alpha*.
- **Chapter 5: Conclusions and Future Work** synthesizes the findings, discusses the limitations of the study, and outlines potential applications in the field of elderly care.

2 State of the Art

The evolution of Conversational Agents has been intrinsically linked to the advancements in NLP and AI. Together, these help to bridge the gap between human communication and machine comprehension. As noted by [15], the implementation of a chatbot system relies on the synergy between AI and NLP to analyze user input and generate responses that mimic human conversation. This process involves multiple layers of processing, from initial pattern recognition to the semantic interpretation of the user's intent, aiming to provide a seamless and intuitive user experience.

Historically, chatbots were limited by rigid rule-based frameworks. However, the field has transitioned toward more sophisticated, contextual architectures. Modern AI assistants are no longer mere responders to isolated commands; they are designed to maintain the flow of dialogue and understand the broader circumstances of the interaction. According to [16], the development of contextual AI assistants represents a significant leap forward, as these systems utilize specialized NLU and core logic units to manage state and context. This allows the assistant to handle complex task analysis and provide emotional support, moving beyond functional utility toward a more human-centric interaction. In recent years, the emergence of GAI and LLMs has further redefined this landscape. While early assistants focused on intent classification and slot filling, current generative models possess the capacity to synthesize vast amounts of information and generate highly coherent, context-aware discourse. This technological shift provides the foundation for more empathetic and deep interactions, which are particularly relevant when addressing specific populations, such as the elderly, where the nuances of language and personal narrative are paramount.

2.1 Generative AI in Assistants for the Elderly

AI for the elderly has shifted from simple monitoring systems to proactive, supportive companions. This transition has been driven by the urgent need to address the challenges of aging populations while enabling independence and improving quality of life. According to [17], the integration of AI-driven technologies and robotics in healthcare serves as a critical facilitator for elderly independence, providing not only physical assistance but also motional support.

The success of these technologies, however, depends heavily on their adoption and the dynamics of the "Smart Elderly Care" ecosystem. As explored in [18], the adoption of AI health assistants is an evolutionary process influenced by the perceived value and the strategic alignment between users and care providers. This research suggests that for an assistant to be truly integrated into the lives of the elderly, it must transcend functional utility and become a trusted component of daily routine, a feat that requires high levels of naturalness and communication reliability.

2.2 Generative AI and the Dynamics of Self-Disclosure

The intersection of GAI and human psychology has opened a new frontier in the study of self-disclosure. As GAI agents become increasingly sophisticated in their linguistic capabilities, the traditional barriers to sharing personal information with non-human entities are more diffuse. This phenomenon is central to understanding how digital companions can facilitate emotional support and relational depth.

According to [19], self-disclosure to conversational AI is no longer a peripheral topic but a structured field of study that requires an emergent framework. This comprehensive literature review

indicates that the way users disclose information to AI is influenced by the agent's perceived empathy and its ability to maintain a coherent, context-aware dialogue. This research emphasizes that future directions in NLP must focus on how AI can proactively encourage disclosure to better serve the user's psychological needs, particularly in longitudinal interactions.

A key validation of this interaction model is found in [20]. Its empirical findings suggest a media equality effect in the context of GAI, demonstrating that individuals provide personal information to AI and humans at equivalent levels. This equivalence is a cornerstone for the present study: if users disclose sensitive information to a chatbot with the same frequency and depth as they would to a human interlocutor, the data extracted through Social Penetration Theory categories becomes a high-fidelity proxy for true emotional and social state.

Consequently, leveraging LLMs for the extraction of disclosure patterns allows for a non-intrusive yet deeply granular analysis of user behavior. By applying automated scoring rubrics to these interactions, we can map the transition from superficial exchanges to core-layer disclosures, providing a quantifiable measure of the social penetration achieved by the AI assistant in the elderly care domain.

2.3 LLM-as-a-Judge

2.3.1 Prompting Strategies: Unified Multitask vs. Fragmented Single-Task Inference

The architectural design of the prompt is a determinant factor in the performance of LLMs (LLMs) acting as evaluators. Two primary strategies were contrasted: a *fragmented approach*, where the model is prompted to provide a score for each individual dimension in isolation (Single-Task), and a *unified approach*, where the model assesses all categories of the SPT within a single inference cycle (Multitask).

The choice between these methodologies is a subject of ongoing debate in the field. According to [21], the comparative analysis of prompt strategies suggests that while single-task prompts may reduce the immediate cognitive load per request, multitask prompting can leverage the model's internal attention mechanisms to account for interdependencies between variables [22].

2.3.2 Granularity of Evaluation Rubrics

The choice of grading scale granularity involves a critical trade-off between sensitivity and reliability. Binary rubrics (0–1) are documented as the most robust against LLM biases, such as verbosity, by establishing clear occurrence thresholds without the risk of false precision. Ternary scales (0–2) offer higher resolution but often face challenges regarding the consistency of the intermediate category boundaries.

Recent research identifies the 0–5 Likert scale as the optimal granularity for human-LLM alignment. According to [23], while models can process wider ranges (e.g., 0–10 or 0–100), these expanded scales introduce noise that decreases the Intraclass Correlation Coefficient (ICC). The 5-point scale represents an empirical "sweet spot," providing sufficient freedom to capture qualitative nuances while maintaining high discriminative stability and alignment with human judgment [23].

2.4 Aggregation Strategies for Subtopic Fusion

Aggregating granular subtopic intensities into high-level thematic dimensions requires the application of mathematical combination rules. The mean operator is the most common approach for capturing the central tendency of a session. It provides a smoothed representation of thematic presence but, as noted in multi-label classification studies, it tends to dilute "peak" emotional events in sparse datasets. Conversely, the maximum operator focuses on the highest intensity detected, preserving significant disclosure events. However, this rule is highly sensitive to outliers and potential LLM hallucinations or over-sensitizations.

The Hadamard product (element-wise multiplication) is increasingly used as a *gating mechanism* to ensure robustness. By multiplying a binary detection vector by a continuous intensity vector, it effectively filters out noise. According to the foundational framework in [24], while the sum rule (Mean) is more resilient to noise in probability estimation, the product rule (Hadamard) is superior when the different sources (subtopics) provide independent and complementary evidence. In the context of automated text analysis, this gating strategy ensures that intensity scores are only considered valid if a prior binary threshold of thematic presence is met.

3 Methodology and Experimental Design

3.1 Preliminary Subtopic Clustering Trials

Before deciding the final the top-down categorical framework, an exploratory bottom-up clustering analysis was performed on the granular subtopics. The objective was to determine if these subtopics would naturally aggregate into distinct, non-overlapping thematic clusters that could independently validate the proposed Social Penetration Theory dimensions.

The preliminary clustering was executed by treating each subtopic's intensity as a coordinate in a high-dimensional space. Upon examining the resulting centroids, it was observed that each cluster center was typically dominated by a specific subtopic belonging to one of the main categories. However, despite this individual identification, the overall boundaries between the clusters remained mathematically indistinct, showing high levels of overlap and low silhouette coefficients.

The failure to identify natural groupings through unsupervised clustering can be attributed to the high thematic volatility inherent in natural conversation. Unlike structured text, elderly-bot interactions are characterized by rapid transitions; a user may shift from a biographical anecdote (*History*) to a present health concern (*Health*) or a mention of loneliness (*Relationships*) within a very short temporal window.

This semantic drift implies that subtopics do not exist in isolated silos throughout a session. Instead, they are interwoven in a way that defies traditional partitioning algorithms. Because the conversation evolves rapidly and topics are often functionally unrelated in the short term, the resulting feature space is too sparse and noisy for a purely bottom-up approach to yield meaningful categories. This finding reinforced the necessity of adopting a theoretically grounded framework (the 3 SPT categories) as an anchor to provide the semantic structure that the raw data, by its conversational nature, does not provide spontaneously.

3.2 Methodological Research Axes

The semantic characterization of the corpus was restructured into three main research axes setups, regarding the architecture of the prompt, the granularity of the numerical scale, and the alignment of the mathematical aggregation of subtopics with human evaluators.

Table 1 outlines these as high-level objectives, focusing on the trade-offs between prompt specificity, scoring complexity, and data fusion techniques.

Table 1: Research Axes and Methodological Goals.

Research Axis	Objective	Focus
Axis I: Architecture	Analysis of Prompt Scope.	Individualized vs. Unified prompts.
Axis II: Granularity	Evaluation of Scoring Resolution.	Binary vs. Rubric (0-5).
Axis III: Fusion	Impact of Aggregation Functions.	Mean, Max, and Hadamard operators.

To facilitate the mapping between the computational implementation and the statistical analysis, the following nomenclature was defined for the three variants. These variants represent specific combinations of prompting strategies and scoring resolutions.

Table 2: Technical nomenclature and configuration of the methodological goals.

Phase	ID	Scoring	Fusion	Prompt Strategy
I: Architecture	1	Rubric (0–5)	-	Categories: Individual vs. Unified prompts comparison.
II: Scoring	2	Binary (0–1)	-	Categories: Evaluation under binary scale.
	3	Trinary (0–2)	-	Categories: Evaluation under trinary scale.
	4	Rubric (0–5)	-	Categories: Evaluation under rubric scale.
III: Aggregation	5	Binary / Trinary / Rubric	Mean	Subcategories: Mean aggregation across all scoring schemes.
	6	Binary / Trinary / Rubric	Hadamard	Subcategories: Hadamard fusion across all scoring schemes.
	7	Binary / Trinary / Rubric	Maximum	Subcategories: Max operator across all scoring schemes.

Note on Experimental Design: *While scoring granularity was a primary focus of experiment 2, its re-evaluation within these setups serves as a crucial reinforcement layer. This approach ensures the robustness of the metrics when subjected to varying prompt architectures and fusion mechanisms, cross-validating the consistency of the model's performance across phases.*

Mathematical Definitions for Subtopic Fusion

In Experiment C, the synthesis of sub-dimensional data into a category-level score is governed by three distinct mathematical approaches (entity definitions in Table 3):

1. **Arithmetic Mean (μ):** Provides a balanced representation of the general intensity across all subtopics within a topic.

$$\text{Score}_{cat} = \frac{1}{n} \sum_{i=1}^{n} (s_i \cdot p_{cat,i})$$

2. **Hadamard Product ($\odot$):** Used to integrate presence/absence detection with intensity rubrics, effectively filtering the score based on explicit evidence.

$$\text{Score}_{cat} = \frac{1}{n} \sqrt{\sum_{i=1}^{n} (s_i \cdot p_{cat,i})^2}$$

3. **Maximum Operator** (max)**:** Highlights the most prominent sub-dimension, identifying the peak level of self-disclosure within a specific topic.

Table 3: Mathematical Definitions of Variables for Subtopic Fusion

Variable	Notation	Description
Intensity	s	Raw score assigned by the LLM representing the qualitative depth of disclosure of a subtopic.
Count	n	Total number of sub-dimensions or subtopics associated with a specific macro-category/topic.
Presence	p	Binary indicator ($p \in \{0, 1\}$) where 1 indicates the subtopic is present and 0 indicates absence.
Subtopic Score	$Score_{sub}$	The weighted value of a sub-dimension, typically calculated as $s \times p$.
Category Score	$Score_{cat}$	The final aggregated intensity value for a macro-category (Health, Relationships, or History).

$$\text{Score}_{cat} = \max((s_1 \cdot p_{cat,1}), (s_2 \cdot p_{cat,2}), \ldots, (s_n \cdot p_{cat,n}))$$

3.3 LLM-Based Feature Extraction

First, we begin by defining our core categories (C_1, C_2, C_3) and the specific subtopics that constitute each thematic domain. This structural mapping, as detailed in Listing 1, serves as the organizational framework for our analysis, ensuring that each granular sub-dimension is correctly aligned with its corresponding higher-level Social Penetration Theory category.

```
category_mapping = {
    'Category 1: Health and Well-being': [
        'mental_emotional_health',
        'physical_health_addictions'
    ],
    'Category 2: Interpersonal Relationships': [
        'family_dynamics',
        'affective_relationships'
    ],
    'Category 3: Personal History and Development': [
        'personal_goals_failures',
        'traumas_adversities',
        'professional_career'
    ]}
```

Listing 1: Mapping of Subtopics to SPT Categories.

With these mathematical foundations established, the previously defined vectors can now be populated with their respective numerical values:

$$\mathbf{S} = [s_1, s_2, s_3, s_4, s_5, s_6, s_7] \tag{1}$$

$$\mathbf{P}_1 = \begin{bmatrix} 1\,1\,0\,0\,0\,0\,0 \end{bmatrix}, \quad \mathbf{P}_2 = \begin{bmatrix} 0\,0\,1\,1\,0\,0\,0 \end{bmatrix}, \quad \mathbf{P}_3 = \begin{bmatrix} 0\,0\,0\,0\,1\,1\,1 \end{bmatrix} \tag{2}$$

The organization of the feature extraction experiments is summarized in Table 4, which outlines the specific objectives and methodological protocols for each experimental setup.

Table 4: Summary of LLM-Based Feature Extraction Experiments.

Experiment	Objective	Methodology
4.2.1: Exp. A	Fine-grained analysis of Subtopics.	Rubric vs. Binary evaluation for individual subtopics.
4.2.2: Exp. B	Integration of intensity and presence via Hadamard Product.	Rubric (0-5) $\odot$ Binary (0/1) at subcategory level.
4.2.3: Exp. C	Direct comparison of scoring granularity at the Category level.	Rubric vs. Binary evaluation for C1, C2, and C3.

To ensure consistency between the computational implementation and the analytical results, a standardized nomenclature was established for the six experimental variants, as detailed in Table 5. These are grouped into three main categories based on their granularity (Category vs. Subtopic) and their scoring logic (Rubric, Binary, or Hadamard Fusion).

Table 5: Technical nomenclature of the experiments conducted.

Exp.	ID	Technical Name	Description
A	1	`rubric based hadamard fusion`	Category-level rubric scores.
	2	`binary hadamard fusion`	Category-level scores from binary flags.
B	3	`topic rubric`	Direct evaluation of the 3 main categories using a 0-5 scale.
	4	`topic binary`	Direct presence/absence detection of the 3 main categories.
C	5	`subtopic rubric`	Detailed 0-5 scoring for each specific subtopic.
	6	`subtopic binary`	Presence/absence detection for each specific subtopic.

Experiment B represents our primary approach, where we hypothesize that multiplying the rubric score by binary flag vectors (defined as Presence P) to later apply the Hadamard Product, that filters out low-confidence model assignments while preserving the intensity of high-quality disclosures.
Experiments A and C serve as ablation studies and validation baselines to justify the complexity of the feature engineering process.

3.3.1 Experiment A: Subtopics (Rubric vs Binary)

The experimental workflow follows a bottom-up structure, beginning with Experiment C (Subtopics) due to a critical data dependency. This experiment serves as the foundational feature extraction layer; it is a mathematical prerequisite to obtain granular scores for each specific subdimension before executing higher-level operations.

The evaluation of subtopics was performed with a zero-shot/few-shot hybrid prompt. This prompt defines the semantic boundaries for each of the seven subtopics and establishes a rigorous scoring rubric. This prompt is described in Annex A.1.

For the binary variant of Experiment C, the model's instructions were simplified to a detection task. The prompt explicitly forbids intermediate values, forcing a strictly boolean output (0 or 1). This reduction in dimensionality serves to minimize variance and provide a high-confidence indicator of thematic occurrence (See Appendix A.1).

3.3.2 Experiment B: Rubric-Based vs. Binary Scoring via Hadamard Product

The hadamard product is used to merge the qualitative depth of the scores.

First, the gating mechanism in matrix-vector form is denoted as:

$$\begin{bmatrix} s_1 \\ s_2 \\ \vdots \\ s_n \end{bmatrix} \odot \begin{bmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{bmatrix} = \begin{bmatrix} s_1 \cdot p_1 \\ s_2 \cdot p_2 \\ \vdots \\ s_n \cdot p_n \end{bmatrix} \tag{3}$$

By substituting the components of the Hadamard product ($f_i = s_i \cdot p_i$), the final operational formula for Experiment B becomes:

$$S_{cat} = \frac{1}{n} \sqrt{\sum_{i=1}^{n} (s_i \cdot p_i)^2} \tag{4}$$

3.3.3 Experiment C: Direct Category Evaluation (Rubric vs Binary)

The rubric-based prompt for Experiment C (See Appendix A.2) provides the model with a nuanced scale to differentiate between impersonal mentions and deep emotional vulnerability. This allows for a direct comparison with the aggregated results of Experiment A to determine if "direct inference" leads to a loss of granularity.

The prompt for Experiment C (Binary) (also in Appendix A.2) instructs the model to act as a high-level classifier. Unlike the subtopic prompt, this prompt requires the LLM to perform an internal aggregation, considering various aspects (e.g., physical health and emotional state) under a single categorical label.

3.4 Unsupervised Machine Learning

Unsupervised machine learning comprises a set of techniques designed to identify underlying structures and patterns in datasets that lack explicit labels. In the context of this thesis, where the objective is to categorize the depth of 16,000 spontaneous interactions without a pre-existing annotation, this paradigm provides a scalable and objective framework for behavioral discovery.

Following the comprehensive review in [25], unsupervised learning is fundamental for exploratory data analysis, as it allows for the identification of natural groupings based on feature proximity in a multi-dimensional space. The utility of this approach in social computing lies in its ability to handle high-dimensional semantic vectors—such as those extracted via LLMs—and organize them into interpretable clusters without human bias influencing the initial grouping process.

A critical component of unsupervised learning is the evaluation of the resulting data partitions. Since there is no external label to compare against, in [25] internal validation criteria are emphasized to measure clustering quality:

- **Cohesion:** Measures the degree to which elements within a cluster are similar to one another.
- **Separation:** Evaluates the distinctness between different clusters to ensure they represent unique categories.
- **Validation Metrics:** Indicators such as the *Silhouette Score* (measuring point-to-cluster assignment quality) and the *Davies-Bouldin Index* (evaluating the ratio of within-cluster to between-cluster distances) assess the mathematical validity of the model.

This theoretical grounding makes clustering a suitable tool for mapping conversational data to the layers of SPT, providing the necessary metrics to evaluate whether the extracted features are capable of forming stable and meaningful representations of user behavior.

3.4.1 K-Means Clustering

K-Means is an unsupervised learning technique to partition n observations into K clusters, where each observation belongs to the group with the nearest mean (centroid). The mathematical objective is to minimize the *Within-Cluster Sum of Squares* (WCSS):

$$J = \sum_{j=1}^{K} \sum_{x_i \in C_j} \|x_i - \mu_j\|^2 \tag{5}$$

To evaluate the validity of the partition and select the optimal number of clusters, this study employs the following metrics and heuristics based on [25]:

- **Elbow Method:** Identification of the inflection point in the WCSS curve.
- **Silhouette Score:** A measure of cohesion and separation ranging from -1 to 1.

3.4.2 Dimensionality Reduction and Visualization

The feature vectors extracted are processed within their original high-dimensional latent space to ensure the theoretical integrity of the SPT categories. To avoid the loss of granular information, K-Means clustering is applied directly to the extracted features without prior reduction.

Table 6: Summary of Clustering Evaluation Metrics and Optimal Selection Criteria.

Metric / Method	Optimal Value Criterion
Elbow Method	”Knee” or Inflection point
Silhouette Score	Maximize ($\rightarrow 1$)

Consequently, dimensionality reduction techniques are used strictly for *post-hoc* analysis and visualization.

Principal Component Analysis (PCA) is implemented as a deterministic validation tool. By projecting the dataset into its primary eigenvectors, PCA allows for the inspection of the global data structure and the verification of cluster separability based on maximum variance.

Furthermore, t-Distributed Stochastic Neighbor Embedding (t-SNE) is utilized to represent complex, non-linear local relationships in a 2D plane. We remark that neither PCA nor t-SNE are used to feed the clustering algorithm; their role is to provide a human-interpretable validation of the high-dimensional mathematical partitions, ensuring they reflect coherent interaction patterns.

Table 7: Comparison between PCA and t-SNE for Dimensionality Reduction.

Feature	PCA	t-SNE
Type	Linear	Non-linear
Data Preservation	Global structure (Variance)	Local structure (Similarity)
Computational Cost	Low / Efficient	High / Iterative
Determinism	Deterministic	Stochastic (Non-deterministic)
Output	Principal Components	Low-dimensional clusters
Main Use Case	Noise reduction / Processing	Data visualization

3.4.3 Implementation of the K-Means Algorithm

The clustering phase was executed using the K-Means algorithm, chosen for its interpretability in centroid analysis. The model was applied directly to the feature space to maintain the SPT. While the Silhouette Coefficient and Elbow Method results were not entirely decisive (suggesting a continuous distribution of user behaviors), the algorithm successfully identified core trends in the data.

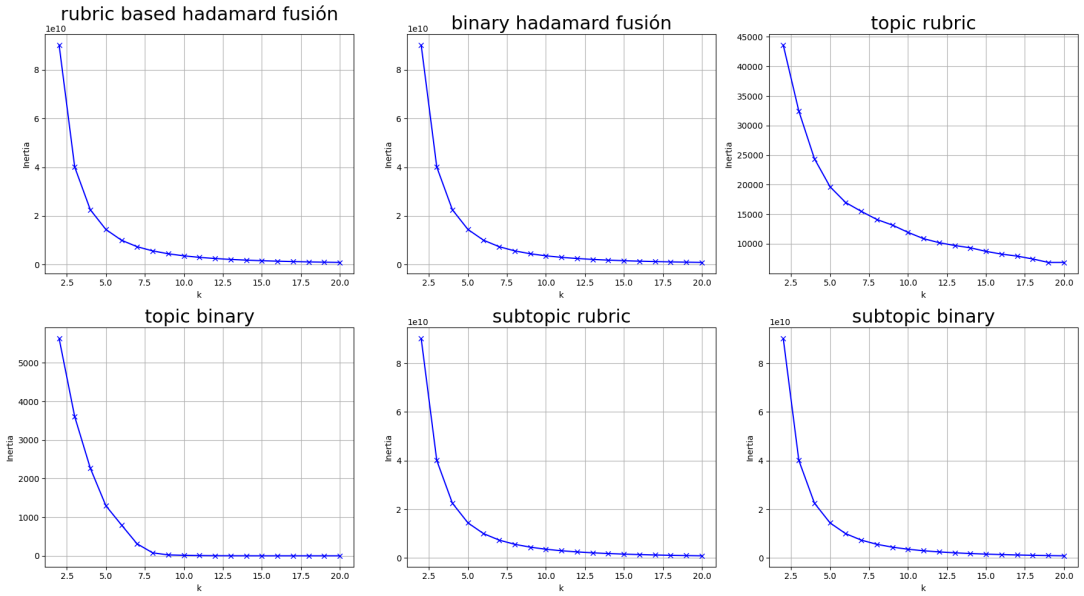


Figure 1: Elbow Method analysis for cluster optimization.

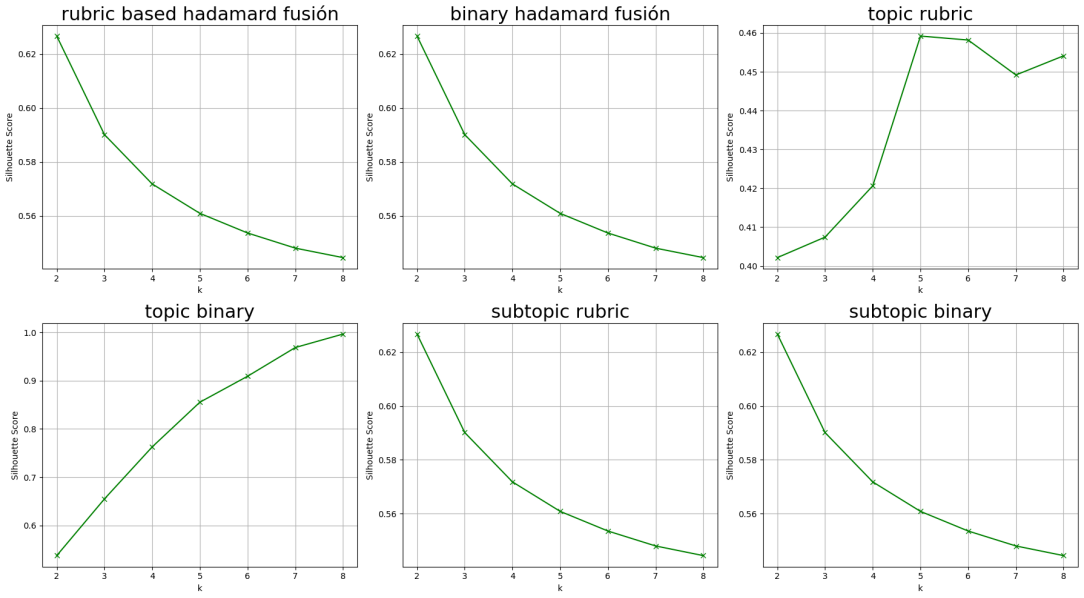


Figure 2: Silhouette Method analysis for cluster optimization.

As illustrated in Figure 1, the inertia curve does not exhibit a clearly defined ”elbow” or a sharp inflection point. This lack of a decisive break suggests that the self-disclosure patterns in our elderly-chatbot dataset do not form naturally isolated groups with high density. Instead, the results point towards a behavioral continuum, where the boundaries between different levels of intimacy are fluid and overlapping.

Despite the absence of a mathematically unique optimal k, the selection of an *N+1* approach prioritizes qualitative interpretability over purely mathematical partition metrics. Specifically, we propose a 3+1 cluster approach. The 3 clusters represent the stable behavioral archetypes identified through inference (and they would target the aforementioned C_1, C_2 and C_3). The additional cluster is strategically designed to capture mixed interactions. This separation is crucial in elderly-bot datasets to prevent “cold-start” sessions from diluting the semantic clarity of the

primary topic centroids.

4 Results and Validation

4.1 Characterization of the experimental setup

4.1.1 Ethics and Privacy in Sensitive Data Analysis

The analysis of natural language in interactions with elderly users presents significant challenges regarding data privacy and ethical handling. This study implements a ”Privacy by Design” approach to ensure compliance with security standards and the protection of sensitive information.

- **Local Infrastructure and Data Sovereignty:** To ensure maximum compliance with the General Data Protection Regulation (GDPR) and the principle of ”Privacy by Design” (Art. 25 GDPR), this research bypasses third-party Cloud-based LLM providers. Instead, a fully local inference pipeline was established.

  As a reference LLM, y deploying the *Gemma-2-9b-it* model on an internal NVIDIA A10 server with the Text Generation Inference (TGI) framework, the study maintains absolute data sovereignty [26]. This architecture ensures that raw, sensitive conversational data from elderly users never leaves the institutional network, thereby eliminating risks associated with international data transfers or the terms of service of external proprietary APIs.

  From a technical perspective, the use of TGI on an A10 GPU (24GB VRAM) allows for high-throughput execution and optimized memory management (e.g., through continuous batching), which significantly reduces latency. Furthermore, this local setup grants full auditability from ingestion to Inference, providing a level of transparency and control that cloud-based solutions cannot guarantee.

- **Data Anonymization and Traceability:** The dataset is structured around unique session and user identifiers (UUIDs), devoid of any direct personal identifiers such as names or contact information. Although no automated Named Entity Recognition (NER) preprocessing was applied to the raw text, the risk of re-identification is mitigated by the restricted access to the dataset and the fact that metadata is limited to technical variables (e.g., timestamps, platform type, and game modes).

This framework ensures that the pursuit of technical innovation in social computing does not compromise the fundamental rights of vulnerable demographics.

4.1.2 Data Source and preproccesing: 16,301 Conversations with Elderly Users

Serenia Solutions (https://sereniasolutions.com/) is a startup specialized in the development of virtual companions designed to alleviate the solitude of the elderly through conversational AI. Its ecosystem enables a wide variety of *interactions* between the user and a chatbot, ranging from affective support to entertainment through *games* and ludic activities. For the purposes of this study, the platform provided the ideal environment to observe how self-disclosure manifests in a controlled digital context.

The initial stage of this research involved the curation of a large-scale conversational corpus of Serenia interactions. The raw data comprised approximately 30,561 sessions. However, to

ensure the reliability of the self-disclosure analysis, a multi-stage preprocessing pipeline was applied to refine the dataset.

1. **Filtering of Ludic Interactions:** Messages categorized under the `intent == games` label (e.g., riddles, word searches) were removed. These interactions are governed by rigid game logic rather than spontaneous social exchange, making them unsuitable for studying natural self-revelation.

2. **System Message Cleaning:** Technical artifacts, such as the automated response `(pensando...)` generated during system latency, were purged. This ensures that the subsequent NLP tasks are performed strictly on human and intentional bot-generated content.

3. **Minimal Interaction Constraint:** A threshold was set to exclude sessions with fewer than five total exchanges (human-bot turns). This criterion ensures that only sessions with enough conversational ”depth” are included in the study.

As a result of this process, the final dataset was reduced to 16,301 valid sessions corresponding to 2,174 different users. For data management purposes, each session was mapped from a composite identifier `'{user_id}_{session_id}.csv'`. This structured approach allows for efficient feature engineering while maintaining the integrity of the original conversational flows.

The relationship between users and interactions provides insight into the engagement patterns of the elderly population with the AI system.

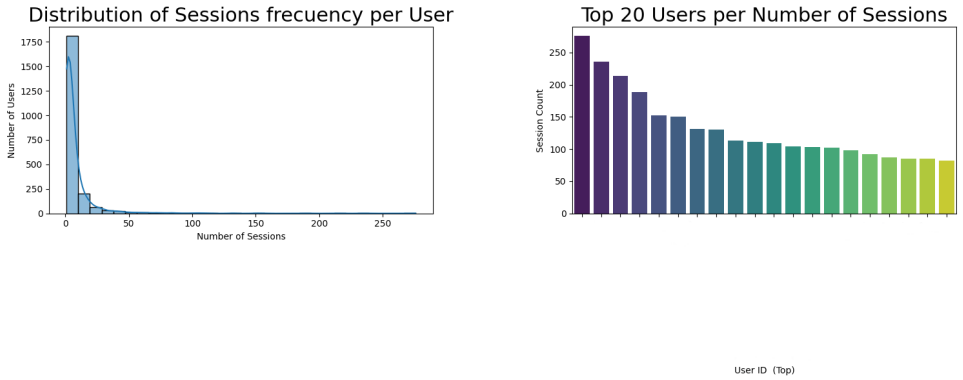


Figure 3: Distribution of the number of conversations per user (Top 20 most active users).

As shown in Figure 3, there is a significant variance in user activity. The concentration of data in the most active users allows for a robust analysis of conversational depth, as these long-term interactions are more likely to transition from superficial *cliché* exchanges to more intimate layers of disclosure.

4.2 Fiability and Quality Standard

To validate the reliability of the LLM-based feature extraction (using *Gemma-2-9b-it* as previously mentioned), and provide a benchmark for performance evaluation, a Gold Standard dataset was established. This ground truth consists of 130 carefully selected conversational sessions, manually annotated by human experts to ensure a representative distribution across the different Social Penetration Theory dimensions and intensity levels.

4.2.1 Selection Strategy and Sampling Criteria

The 130 sessions were divided into two primary groups based on their preliminary scoring distributions to ensure that the human evaluators encountered both unambiguous and borderline cases:

- **High-Intensity Group (80 sessions):** This group aimed to capture the most distinct examples of self-disclosure. We selected 20 sessions for each of the three primary categories to which a LLM assigned the maximum intensity score. Additionally, 20 sessions were selected for the non-disclosure baseline, defined as cases where the scores across all categories were strictly lower than 2 ($S_c < 2$).
- **Distributional Refinement Group (50 sessions):** This group was curated to test the model's discriminative precision.
  - **Predominant Category (30 sessions):** 10 sessions per category were chosen under the condition that one specific category had an intensity score > 3 and was significantly higher than the scores of the remaining categories.
  - **Intermediate Disclosure (20 sessions):** Two subgroups of 10 sessions each were selected based on intermediate scoring ranges to evaluate the model's sensitivity to subtle nuances in the user's narrative.

4.2.2 Human Annotation Protocol

The labeling process was carried out by a panel of three researchers in the field. To eliminate bias and ensure the integrity of the Gold Standard, the following protocol was implemented:

1. **Blind Evaluation:** Annotators were not provided with the scores previously generated by the reference LLM model.
2. **Scoring Metric:** Each session was manually evaluated by every annotator using a standardized scale from 0 to 5, matching the rubric used in the categories prompt (See Appendix A.2).
3. **Independent Review:** Evaluators worked independently to assess the presence and intensity of disclosure in each of the three theoretical categories.

This curated dataset serves as the definitive reference for calculating the error margins, providing a statistically sound foundation for the subsequent analysis of results.

4.2.3 Validation of the gold standard: Interagreement

To assess the reliability of the human-annotated benchmark, we calculated the Cohen's Kappa coefficient across all pairs of evaluators for the three primary categories. The results, illustrated in Figure 4, show agreement levels ranging from 0.32 to 0.52, which conventionally corresponds to "fair" to "moderate" agreement.

While these values may appear numerically low, they are expected and justified within the context of natural language processing for sensitive populations for the following reasons:

- **Subjectivity of Intimacy:** Interpreting "depth" in elderly self-disclosure is inherently non-binary; a score of 3 versus 4 often depends on individual evaluator perception of vulnerability.

- **Thematic Overlap:** SPT categories are fluid; complex disclosures (e.g., family loss) frequently trigger multiple categories (C_2 and C_3), fragmenting agreement.
- **Metric Sensitivity:** Cohen's Kappa penalizes any 1-point difference on a 6-level scale (0-5) as a total disagreement, even when annotators agree on the general presence and intensity of a topic.

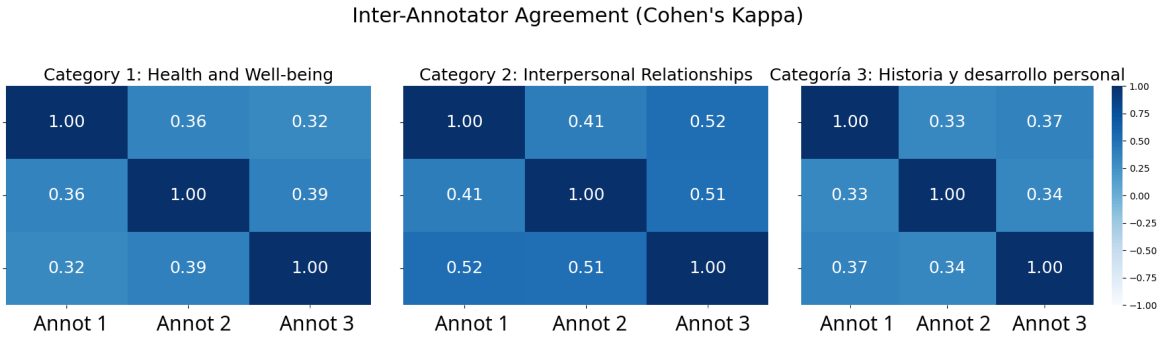


Figure 4: Inter-Annotator Agreement (Cohen's Kappa) across the three primary categories.

Therefore, we applied Krippendorff's Alpha (α), Mean Point Distance, and Spearman Correlation (ρ) to assess the "distance" between human ratings. Table 8 summarizes the consistency across the three primary categories.

Table 8: Consolidated Reliability Metrics for the Gold Standard Dataset.

Category	**Krippendorff's α**	**Avg. Point Distance**	**Mean Spearman ρ**
C_1: Health and Well-being	0.6634	0.96	0.70
C_2: Interpersonal Relationships	0.8311	0.66	0.83
C_3: Personal History	0.6698	0.80	0.69

The analysis reveals several insights into the quality of the human-labeled benchmark:

- **High Consistency in C_2:** Interpersonal Relationships achieved the highest reliability ($\alpha = 0.83$), suggesting that experts find social dynamics easier to quantify than health or historical nuances.
- **Sub-Point Error Margin:** Across all categories, the average discrepancy between any two evaluators is consistently below 1.0 points on the 0-5 scale. This indicates that even when annotators disagree, they remain within a close numerical range, validating the stability of the intensity rubric.
- **Strong Linear Trends:** The high Spearman Correlation values (averaging $\sim$0.74) confirm that evaluators identify the same relative peaks and valleys of self-disclosure, even if their absolute baseline scores vary slightly.

In conclusion, these results confirm that the Gold Standard is a robust reference. The low average distance (< 1.0) suggests that the experimental results (Exp. A, B, and C) can be evaluated with a high degree of confidence, as the ground truth provided by humans is mathematically consistent despite its inherent subjectivity.

4.3 Evaluation of the Metodological Axes

4.3.1 Impact of Prompt Architecture on Extraction Accuracy

A comparative analysis of the prompt engineering strategies reveals that consolidated inference outperforms fragmented requests (See Table 2: Architecture). As illustrated in Figure 5, the single-prompt approach exhibits a highly favorable error distribution, with a significant concentration of errors at the 0 and 1 levels. In contrast, the multi-prompt architecture flattens this curve, showing a concerning increase in higher-magnitude errors (3 to 5 range), suggesting that isolating categories may lead to "intensity drift" or over-sensitization by the LLM.

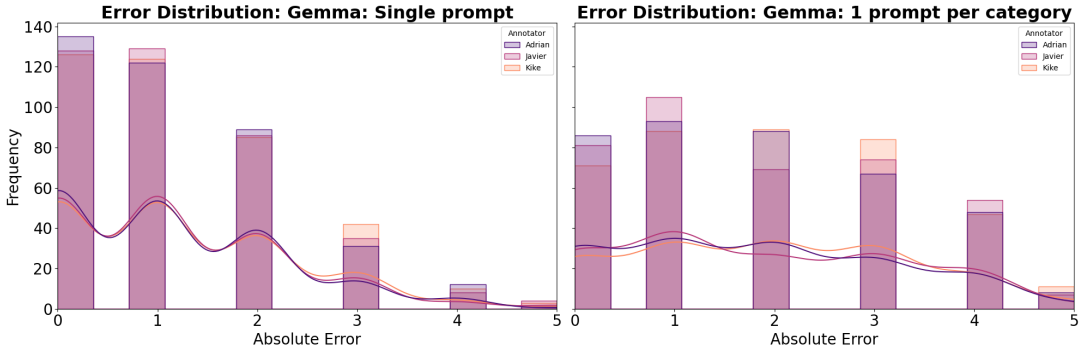


Figure 5: Error dispersion comparison: single prompt vs one prompt per category.

This trend is quantitatively validated by the Mean Absolute Error (MAE) benchmarks shown in Figure 6. The data reveals two key insights:

- **Categorical Consistency (Figure 6a):** The single-prompt model achieves a lower MAE across all three theoretical dimensions. Notably, in the *Interpersonal Relationships* category, the error drops below 1.0, whereas the multi-prompt setup consistently remains above the 1.5 threshold.
- **Annotator Alignment (Figure 6b):** The single-prompt architecture maintains a stable error margin (≈ 1.2) regardless of the human evaluator, whereas the multi-prompt approach introduces higher variance and significantly higher average error (≈ 1.8 to 1.9).

The success of the single-prompt strategy likely stems from contextual coherence. By evaluating all dimensions within a single context window, the model can maintain a relative scale of intensity across categories, preventing the hallucination that occurs when the LLM is forced to find a score for a specific topic in isolation.

4.3.2 Impact of Scale Granularity on Normalized Error

To evaluate the precision of the LLM against the human Gold Standard, we normalized the error across three experimental configurations: Exp_5 (0–5 scale), Exp_3 (0–3 scale), and Exp_2 (binary/simplified). The results indicate that scale complexity is a primary driver of inferential variance. (See Table 2: Granularity)

- **Optimal Performance in Simplified Scales:** As shown in the heatmap (Figure 7), the Exp_2 configuration achieves the lowest Global Mean Error (0.216). This suggests that while high-granularity scales (0–5) capture more nuance, they do not significantly introduce more noise into the LLM's decision-making process.



(a) MAE by category

(b) MAE by annotator

Figure 6: Comparison of Mean Absolute Error (MAE) across categories and annotators.

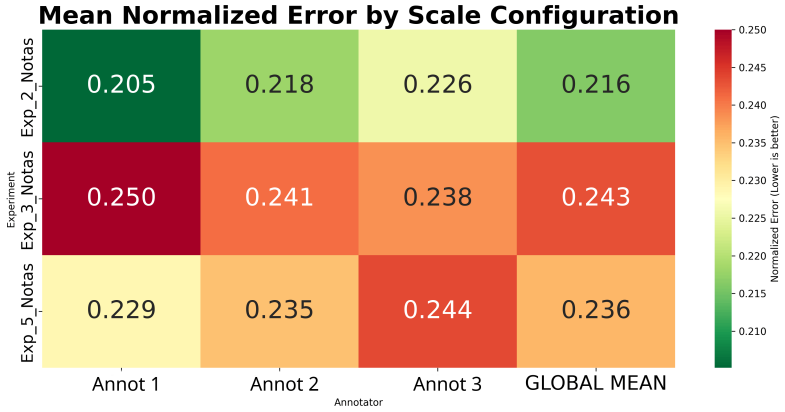


Figure 7: Mean normalized error (0–1) by experiment and annotator. Lower values indicate better performance.

- **Error Stability across Categories:** Figure 8 illustrates that while error is generally suppressed in Exp_2, the *Personal History* category exhibits a higher error ceiling. This indicates that complex, narrative-heavy disclosures are less amenable to drastic scale reduction compared to more objective topics like health.

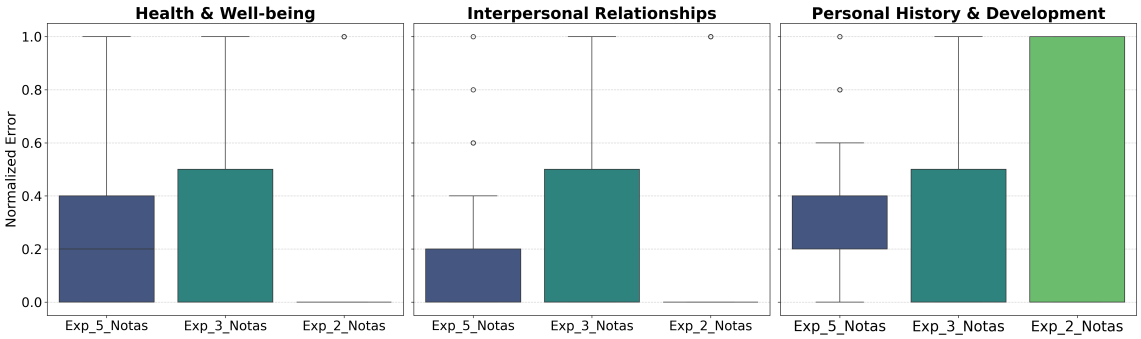


Figure 8: Normalized error distribution by category across different scale configurations.

- **Annotator Invariance:** The consistent error distribution across the three annotators in Figure 9 confirms that the model's performance is not biased toward any specific human evaluator's style. The LLM maintains a systematic error profile, which is crucial for large-scale data processing where human consistency naturally fluctuates.

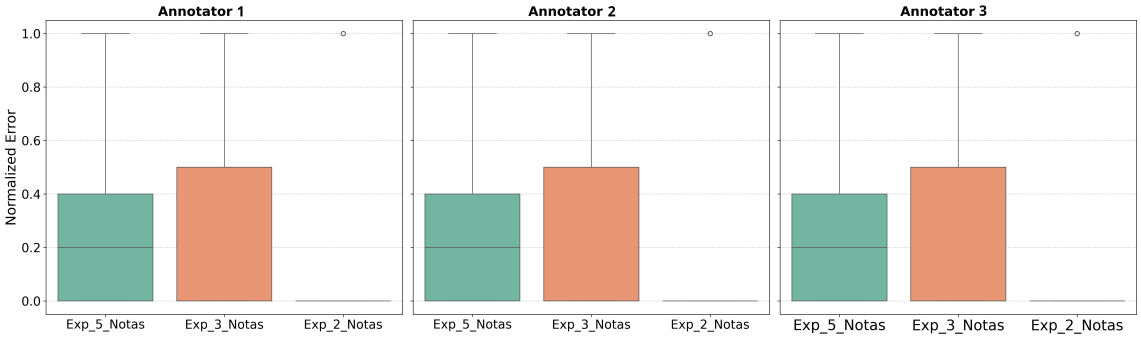


Figure 9: Normalized error distribution by anotator across different scale configurations .

Ultimately, these findings justify the use of Exp_5 for qualitative depth, but also highlight that for tasks requiring high statistical reliability, a reduction in scale granularity (toward the Exp_2 model) significantly stabilizes the output.

4.3.3 Comparative Analysis of Fusion and Aggregation Strategies

The results from the final experimental axis (See Table 2: Granularity), synthesized in figures 11 and 10, confirm that the mathematical method used to combine sub-topic scores is as critical as the prompt structure itself.

- **Superiority of the Gating Mechanism (Hadamard):** The data reveals that the Hadamard Product product architecture consistently yields the lowest error margins. As seen in Figure 11, the "Had" experiments (Exp2, Exp5, and Exp8) outperform the "Mean" and "Max" counterparts across every scale (Formula 1). Specifically, Exp2_Bin_Had achieves the lowest global normalized error (0.196), proving that binary gating effectively filters out hallucinated intensity by requiring explicit topic confirmation.
- **Distribution and Tail Stability:** Figure 10 provides a granular view of this stability. In the "Had" rows, the absolute error distributions show a higher concentration at the zero-error mark and a significantly "thinner" tail compared to "Mean" models. This suggests that Mean fusion tends to "dilute" significant disclosures, while Hadamard gating ensures that intensity is only recorded when the semantic presence is strictly validated.
- **The Depth-Reliability Trade-off:** While Exp2 (Binary) is technically the most precise, Exp8_5R_Had maintains a nearly identical error profile (0.199) while providing a 0–5 granularity. This justifies the selection of the 5-level rubric combined with Hadamard gating as the optimal architecture for capturing conversational depth.

Ultimately, these findings indicate that a gated architecture acts as a robust semantic filter, bridging the gap between high-granularity qualitative data and quantitative statistical reliability.

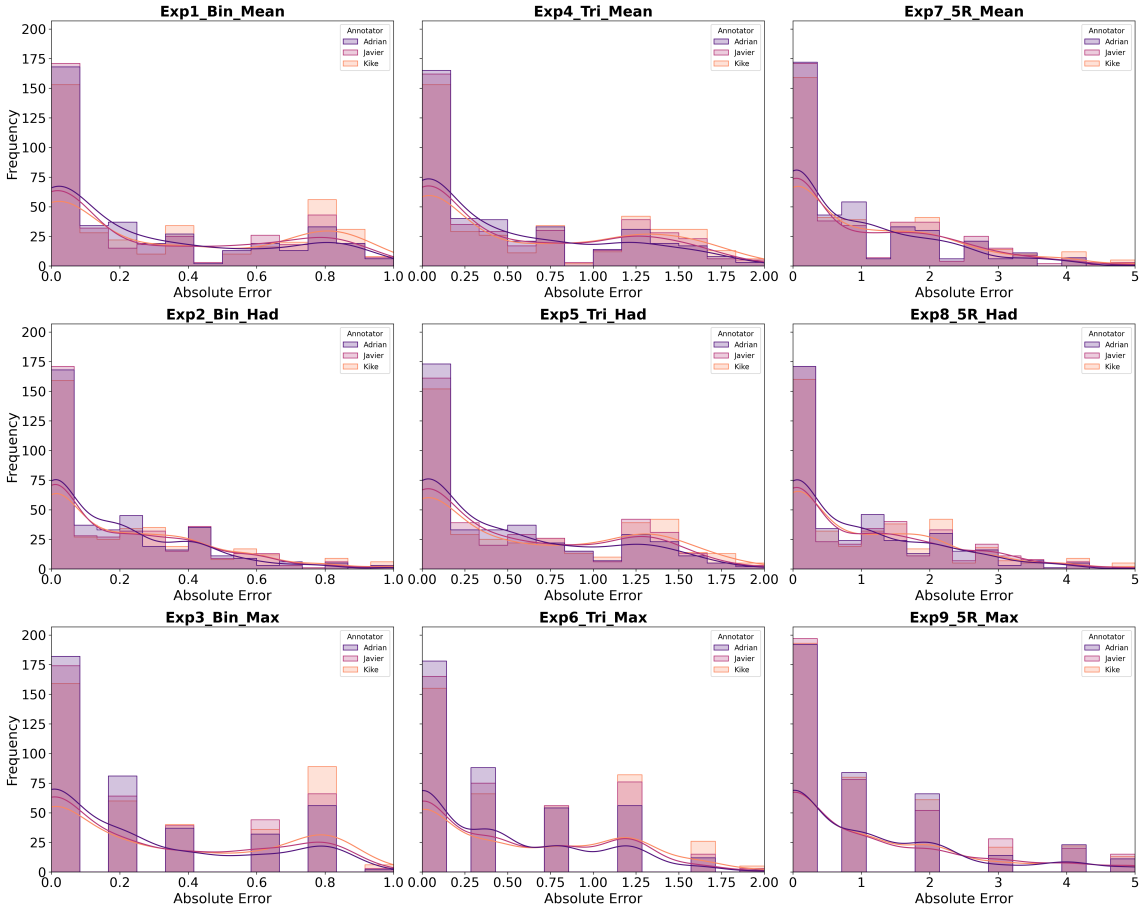


Figure 10: Absolute error distributions across all scale configurations and annotators (3×3 layout).

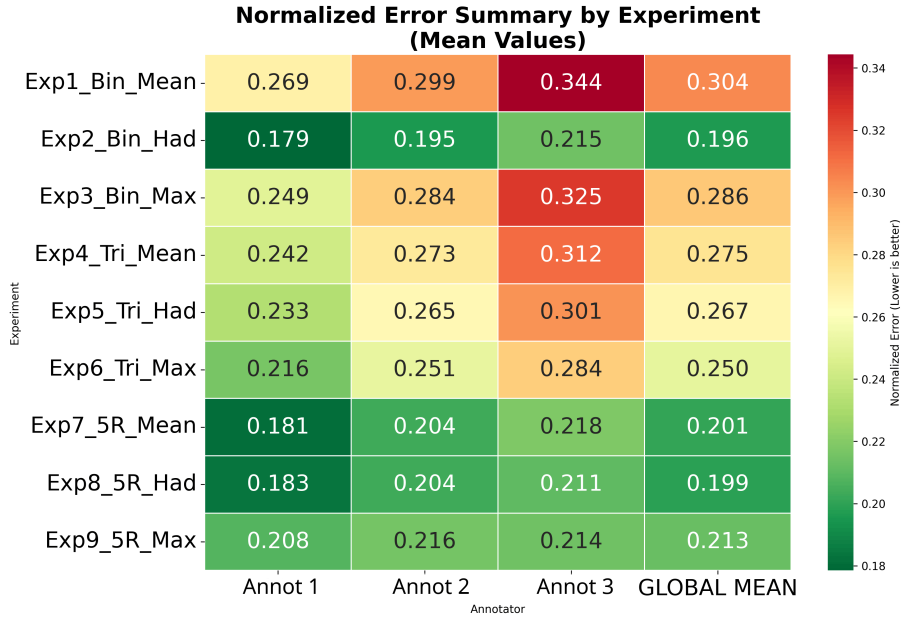


Figure 11: Mean normalized error by experiment and annotator. Lower values indicate better performance .

4.4 Centroid Definition and Cluster Population Dynamics

The analysis of the clustering results across the different experimental setups (those in Table 5) reveals a consistent inverse relationship between centroid definition and cluster size. As observed in Figures 13, 16, and 17, the clusters characterized by highly defined centroids, representing high-intensity, specific self-disclosure, tend to have smaller populations. This suggests that while intense, thematic disclosure exists, it is a particular behavior within the elderly-chatbot interaction space.

This dynamic is further evidenced by the following patterns:

- **Residual Absorption:** Sessions that do not exhibit a clear thematic focus or high intensity are naturally sequestered into the residual cluster ($+1$ cluster). This group acts as a semantic sink for interactions that are either superficial (cliché-level) or highly fragmented, preventing them from diluting the purity of the three primary archetypes.
- **Resolution vs. Fragmentation:** In contrast, less robust configurations (Figure 12, 14 and 15) result in more diffuse, less defined clusters. In those cases, the lack of a strong gating signal causes ambiguous sessions to bleed into the primary categories, reducing the qualitative interpretability of the results.

Ultimately, these results validate the $3+1$ architecture. The concentration of noise into a single, non-specific cluster allows for a much more precise characterization of the three core behavioral profiles, proving that the methodological refinements introduced in the extraction phase are essential for identifying true patterns of social penetration.

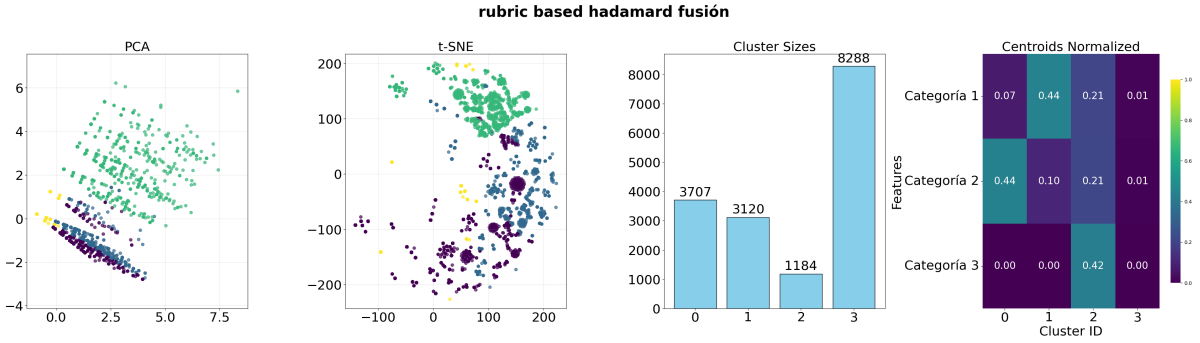


Figure 12: Rubric Based Hadamard Fusion experiment.

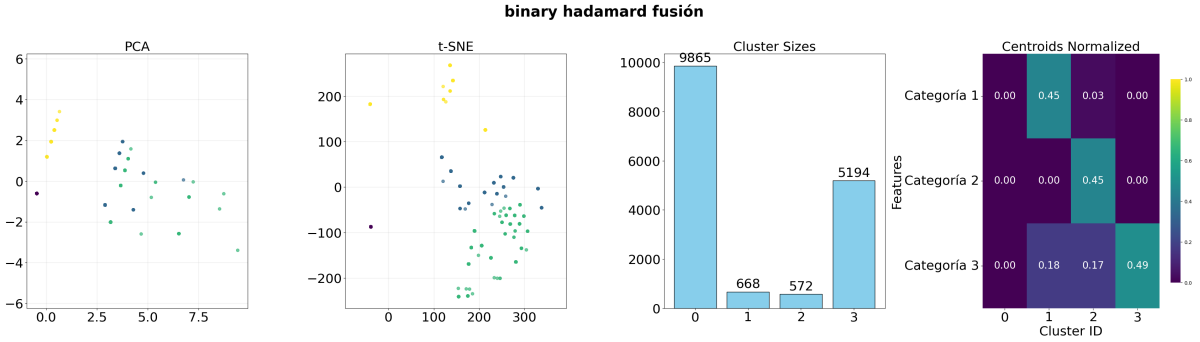


Figure 13: Binary Hadamard Fusion experiment.

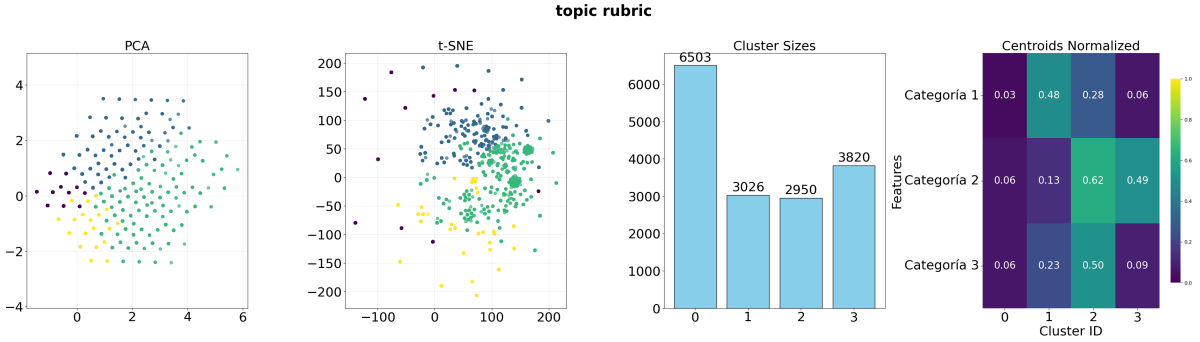


Figure 14: Topic Rubric analysis.

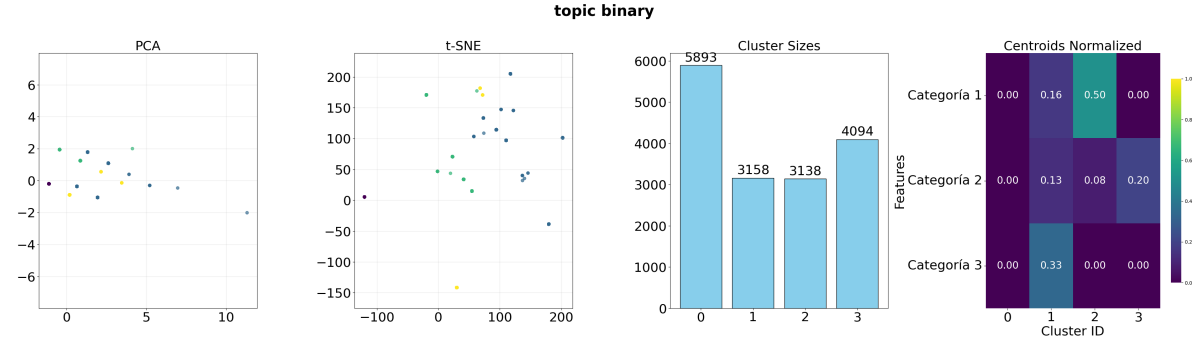


Figure 15: Topic Binary analysis.

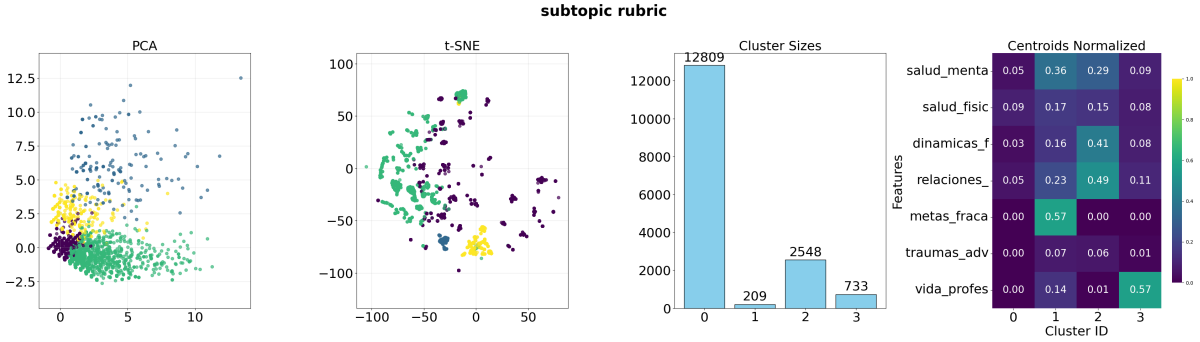


Figure 16: Subtopic Rubric results.

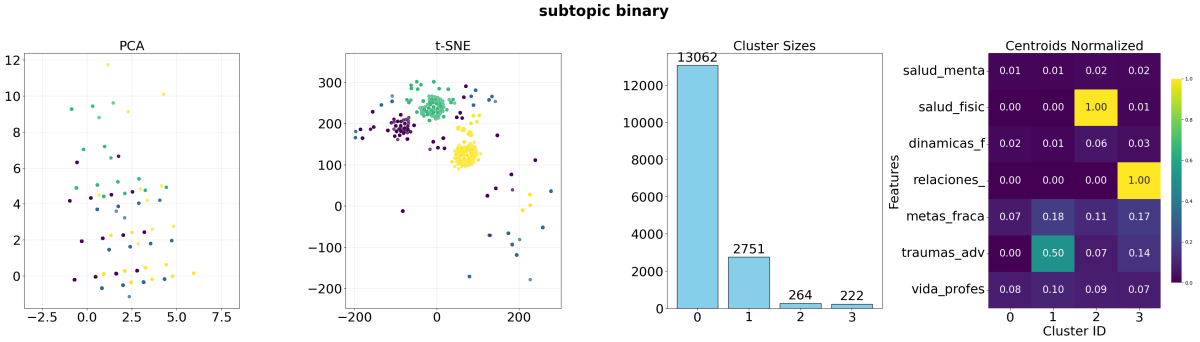


Figure 17: Subtopic Binary results.

Beyond the characterization of thematic clusters, a comprehensive review combining automated intensity scores and manual qualitative spot-checks suggests that approximately one-third of the final dataset (roughly 5,400 out of 16,301 sessions) constitutes true non-disclosure interactions. These sessions are primarily composed of phatic communication, greetings, or purely functional exchanges where the user does not progress beyond the outermost "cliché" layer of the SPT.

The identification of this significant volume of low-intimacy data provides a final empirical justification for our $3 + 1$ clustering strategy. By allowing the residual cluster to absorb this large proportion of non-revealing sessions, the architecture prevents the dilution of the primary behavioral profiles (C_1, C_2, and C_3). This ensures that the identified patterns of health, interpersonal, and historical disclosure remain statistically robust and qualitatively representative of genuine intimate interactions, rather than being overshadowed by the high frequency of superficial conversational mass.

5 Conclusions and Future Work

5.1 Conclusions

This research systematically evaluated the automatic assessment of self-disclosure in interactions between chatbots and older adults using SPT LLMs. Across multiple methodological dimensions, several insights were obtained regarding the extraction and classification of sensitive conversational data.

- **Optimal Prompt Architecture:** The results demonstrate that a single-prompt configuration is superior to fragmented categorical requests. This approach preserves the global context of the conversation, preventing "intensity drift" and ensuring that the LLM maintains internal consistency for scoring different disclosure levels.
- **The Efficiency of Hadamard Gating:** The implementation of a Hadamard Product product for score fusion proved to be the most effective strategy for noise reduction. By requiring explicit semantic confirmation before assigning an intensity value, this mechanism successfully mitigated the hallucination that often plagues standard Mean or Max aggregation methods.
- **Granularity and Stability:** While binary Hadamard scaling offer the highest statistical precision, the 5-level (0–5) intensity rubric remains valuable for qualitative research. When combined with the aforementioned gating strategy, it provides a stable balance between descriptive depth and quantitative reliability.
- **Structural Validity of the $3 + 1$ Model:** The clustering analysis confirms that conversational profiles naturally tend toward specialized thematic centroids (Health, Relationships, and History). The inclusion of a residual ($+1$) cluster to absorb superficial orientation stage sessions was instrumental in maintaining the purity and interpretability of the primary behavioral archetypes.

5.2 Future Work

To deepen the understanding of digital social computing for vulnerable demographic groups, we propose the following future research directions:

- **Clinical Impact Metric (User Well-being):** Future research should focus on validating the system's effectiveness by correlating the depth of spontaneous disclosure with results from standardized clinical loneliness scales. Success would be defined by a measurable reduction in perceived isolation through longitudinal psychometric assessments, rather than solely by conversational engagement metrics [27].
- **Advanced Clustering and Model Selection:** Beyond K-Means, future research should explore probabilistic models such as Gaussian Mixture Models (GMM) and Bayesian Information Criterion (BIC) to determine the optimal number of clusters dynamically. Incorporating the Calinski-Harabasz index and Davies-Bouldin index will provide more rigorous internal validation for group cohesion.
- **Multi-Model and Multilingual Benchmarking:** Evaluation across a broader spectrum of LLMs (e.g., Llama 3, Mistral, and specialized medical-domain models), as well as across diverse languages, would help validate the cross-cultural universality of the current SPT extraction rubrics.
- **Longitudinal Sentiment-Disclosure Correlation:** A novel research direction involves tracking the evolution of sentiment and self-disclosure over months of interaction, potentially serving as a digital biomarker for loneliness or cognitive health [28].
- **Adaptive Conversational Pacing:** Future systems could implement a closed-loop interaction model in which the chatbot dynamically adjusts its conversational strategy in real time. If the system detects that a user remains within the "orientation" layer (C_0), it could autonomously adopt reciprocal disclosure [29] techniques to encourage deeper social penetration [30].

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A Prompts Used in Feature Extraction

This appendix details the system prompts provided to the *Gemma-2-9b-it* model. The structure of these prompts follows the model's control tag format to ensure a deterministic response in JSON format.

A.1 Subtopic Extraction

Experiment C constitutes the granular foundation of the system. Two variants were used: one based on a rubric (0-5) and a binary version (0/1).

```
<start_of_turn>user
You are a text analysis system. Your goal is to quantify the level of self-disclosure in a predefined list of topics:
- Health (Mental/Physical)
- Relationships (Family/Affective)
- Personal History (Goals, Traumas, Career)

Analyze the [MESSAGES TO EVALUATE] and assign a score (0-5) to EACH topic.
Response must be ONLY a valid JSON object.

# EVALUATION SCALE (0-5)
* 0: No Mention.
* 1: Impersonal Mention (General talk).
* 2: Vague Personal Mention (General identification).
* 3: Specific Action Disclosure (Concrete habits/actions).
* 4: Detailed Disclosure (Sharing reasoning/beliefs).
* 5: Deep Disclosure (Vulnerability, impact, deep emotions).

# EXPECTED JSON FORMAT
{
  "mental_emotional_health": Int,
  "physical_health_addictions": Int,
  "family_dynamics": Int,
  "affective_relationships": Int,
  "personal_goals_failures": Int,
  "traumas_adversities": Int,
  "professional_career": Int
}

[MESSAGES TO EVALUATE]:
<end_of_turn>
```

Listing 2: Prompt for Subtopics rubric (Experiment C).

```
<start_of_turn>user
You are a high-precision text classification system. Your task is to detect the PRESENCE or ABSENCE of specific self-disclosure topics in the provided text.

```

```
Analyze the [MESSAGES TO EVALUATE] and assign a value of 1 if the
   topic is mentioned (regardless of its depth) and 0 if it is not
   mentioned at all.
Your response must be ONLY a valid JSON object.

# TOPICS TO DETECT
- mental_emotional_health
- physical_health_addictions
- family_dynamics
- affective_relationships
- personal_goals_failures
- traumas_adversities
- professional_career

# DETECTION RULES
* 0: There is no trace of the topic in the text.
* 1: The topic is present in the text (even if vaguely mentioned).

# EXPECTED JSON FORMAT
{
  "mental_emotional_health": 0 or 1,
  "physical_health_addictions": 0 or 1,
  "family_dynamics": 0 or 1,
  "affective_relationships": 0 or 1,
  "personal_goals_failures": 0 or 1,
  "traumas_adversities": 0 or 1,
  "professional_career": 0 or 1
}

[MESSAGES TO EVALUATE]:
<end_of_turn>
```

Listing 3: Prompt for Subtopics binary (Experiment C).

A.2 Direct topic Evaluation

In Experiment B, the model performs a direct inference on the three macro-categories of the SPT, bypassing the aggregation of subtopics.

```
<start_of_turn>user
You are a text analysis system. Your goal is to quantify the level of self-disclosure in a predefined list of categories focused on lived experiences and the author's inner world.

Analyze the [MESSAGES TO EVALUATE] and assign a score (0-5) to EACH category.
Response must be ONLY a valid JSON object.

# CATEGORIES TO EVALUATE
* 'Category 1: Health and Well-being': Includes physical, mental, and emotional health.
* 'Category 2: Interpersonal Relationships': Includes family dynamics and affective relationships.
* 'Category 3: Personal History and Development': Includes goals, failures, traumas, and career.

# EVALUATION SCALE (0-5)
* 0: No Mention.
* 1: Impersonal Mention (General discussion, no personal link).
* 2: Vague Personal Mention (General identification or labels).
* 3: Specific Action Disclosure (Concrete habits or specific actions)
* 4: Detailed Disclosure (Reasoning, beliefs, or specific data).
* 5: Deep Disclosure (Vulnerability, fear, or profound personal impact).

# EXPECTED JSON FORMAT
{
    "Category 1: Health and Well-being": Int,
    "Category 2: Interpersonal Relationships": Int,
    "Category 3: Personal History and Development": Int
}

[MESSAGES TO EVALUATE]:
<end_of_turn>
```

Listing 4: Prompt for Topic rubric (Experiment B).

```
<start_of_turn>user
You are a text analysis system. Your goal is to quantify the level of self-disclosure in a predefined list of broad categories:
- Category 1: Health and Well-being (Mental, emotional, and physical health)
- Category 2: Interpersonal Relationships (Family dynamics, affective relationships).
- Category 3: Personal History and Development (Goals, failures, traumas, career).

```

```
Analyze the [MESSAGES TO EVALUATE] and assign a 0 or 1 to EACH
   category.
Response must be ONLY a valid JSON object.

# BINARY EVALUATION SCALE (0-1)
* 0: No Mention or Not Relevant: The topic is not mentioned or not
   linked personally.
* 1: Specific Action/Deep Disclosure: The author reveals a concrete
   habit, action, or reflects deeply on the topic.

# EXPECTED JSON FORMAT
{
    "Category 1: Health and Well-being": 0 or 1,
    "Category 2: Interpersonal Relationships": 0 or 1,
    "Category 3: Personal History and Development": 0 or 1
}

[MESSAGES TO EVALUATE]:
<end_of_turn>
```

Listing 5: Prompt for topic binary (Experiment B).